



Awesense Energy Transition Platform Use Cases

● - demoable implementation

🔓 - demoable & open sourced implementation

Jun 6, 2025



UC01	Coincident Peak and Non-Coincident Peak Analysis	🔓	
UC02	Power Factor Analysis		●
UC03	Phase Balance Analysis		●
UC04	Transformer Capacity Analysis for EV Charging	🔓	
UC05	Reversed Power Flow and PV Capacity Analysis	🔓	
UC06	GIS Grid Connectivity Validation and Correction		●
UC07	Asset Management - Cables and Lines		
UC08	Asset Management - Substations		
UC09	EV Charging and Use of Reserved Capacity Analysis	🔓	
UC10	Connected Loads [kW] vs. Actual Phase Balance Analysis		
UC11	Analysis of Unserved Energy Due to Outage Analysis		
UC12	Grid Temperature Monitoring		●
UC13	Master Circuit Breaker Value vs. Measured Values Analysis	🔓	
UC14	Onsite Generation and Ghost Load Analysis		
UC15	Utility Data Challenge Hub		●
UC16	Registered PV Generation Analysis	🔓	
UC17	Actual vs. Contracted Grid Capacity Analysis	🔓	
UC18	Grid Alerts and Warnings		●
UC19	IoT Health Index and Overview		●
UC20	Unexpected Generation Identification		
UC21	Geospatial Analysis of Grid Alerts and Warnings		●



Awesense Energy Transition Platform Use Cases

● - demoable implementation

🔗 - demoable & open sourced implementation

Jun 6, 2025



UC22	Topological Grid Segmentation			●
UC23	Analysis of Grid Segments and Elements			●
UC24	Analysis of Planned Outage for Assets Upgrades	🔗	●	
UC25	Analysis for Maximizing Value of IoT and Meter Devices			●
UC26	Transformer Loading Analysis	●		
UC27	IoT Sensor Billing	●		●
UC28	Energy Balancing for Revenue Protection			●
UC29	Non-Technical Loss Investigations Management			●
UC30	Suspicious Pattern Identification & Analysis			●
UC31	Grid and Time Series Visualization			●
UC32	Power Outage Visualization			●
UC33	MV Grid Optimization Analysis			
UC34	EV and Appliance Disaggregation			
UC35	Standard Consumption Patterns Creation			
UC36	Phase Identification Analysis			
UC37	Detection of Meter Incorrect Wiring Connection			
UC38	Transformer Load Increase Assessment Tool			
UC39	IPT and ICT Verification for Proper Billing			●
UC40	Phase Imbalance Correction Suggester	●		
UC41	EV Load Growth Forecasting	🔗		
UC42	Meter to Transformer Association Analysis			



Awesense Energy Transition Platform Use Cases



- demoable implementation



- demoable & open sourced implementation

Jun 6, 2025



Jupyter



Power BI



TGI

UC43 Load Forecasting

UC44 Voltage Power Quality Monitoring



UC45 Feeder Voltage Analysis and Visualisation

UC46 Feeder Capacity and Lateral Capacity Analysis

UC47 Dynamic Line Rating Analysis

UC48 AMI-Smart Meter Flag Analysis for Theft Detection



UC49 Time Of Use Rate Analysis

UC50 Work Orders Management



UC51 Wildfire Risk Assessment

UC52 Hosting Capacity Analysis

UC53 Grid Map Publication Tool